

Article

Enhancing Open-Loop Wavefront Prediction in Adaptive Optics through 2D-LSTM Neural Network Implementation

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Abstract: Adaptive optics (AO) is a technique with an important role in image correction on ground-based telescopes through the deployment of specific optical instruments and various control methodologies. The synergy between these instruments and control techniques is paramount for capturing sharper and more accurate images. This technology also plays a crucial role in other applications, including power and information systems, where it compensates for thermal distortion caused by radiation. The integration of neural networks into AO represents a significant step towards achieving optimal image clarity. Leveraging the learning potential of these models, researchers can amplify control strategies to counteract atmospheric distortions effectively. Neural networks in AO not only produce results on par with conventional systems but also proffer benefits in cost-efficiency and streamlined implementation. This study explores the potential of an artificial neural network (ANN) as a nonlinear predictor for open-loop wavefronts. Expanding on prior evidence showing advantages over classic methods, this investigation boosts prediction accuracy through the integration of advanced models and machine learning approaches.

Keywords: adaptive optics; machine learning; prediction

MSC: 78A10; 68T05; 82C32



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1. Introduction

Light that reaches ground-based telescopes is distorted due to atmospheric turbulence. The main goal of adaptive optics (AO) is to correct these distortions and restore the original wavefront, ensuring precise measurements. AO techniques typically use a Shack–Hartmann wavefront sensor (SH-WFS) [1] to measure these atmospheric turbulences. This sensor discerns the tilts of the distorted wavefront. The data collected by the SH-WFS are then relayed to the deformable mirror (DM) [2], a flexible reflective surface responsible for correcting the wavefront. The arrangement of these components, along with the data transfer process from the SH-WFS to the DM, defines the configuration of the AO system. Common configurations include the single-conjugated AO (SCAO), relying on a singular guide star for turbulence characterization, and the multiobject AO (MOAO), which uses several nearby stars as reference points for turbulence reconstruction [3].

However, AO not only is a relevant technology for telescope image correction; there are some other applications such as systems of energy and information. Examples include free-space optical communication (FSO), in which AO can be employed to compensate distortion in the optical signal and improve the quality and reliability of optical communication [4]; laser beam propagation, to mitigate the effects of atmospheric turbulence on the laser

beam [5]; and even medical imaging, a scenario where accurate focusing and targeting are crucial [6].

In recent research works, there has been a noticeable increase in exploring the integration of prediction-based control strategies into adaptive optics (AO) systems. The use of artificial intelligence has proven to be a powerful tool in science. This is due to its great ability to learn features and patterns from a large amount of data. AI is made up of individual processing units called neurons, so it mimics the process of interconnections and learning of a biological neural system.

Taking into account, in addition to this, atmospheric turbulence can be modeled as a stack of static layers that are independent of each other, and that move with a velocity dependent on the wind that influences each of them, as can be seen in the work of [7,8] on the basis of the frozen flow hypothesis. This means that the evolution of this variable is not completely random, so it can be predicted in a certain way based on the partial temporal and spatial dependence that it possesses.

Considering previous points, using a neural network applied to AO, the prediction model takes the information from the wavefront sensor and provides a tomographic reconstruction of the turbulence such that the aberrated wavefront is compensated with a deformable mirror [9].

Those mentioned strategies involve predicting turbulence evolution, allowing control mechanisms to finely adjust the caption of light from celestial objects with a precision similar to conventional real-time control systems (RTCs). Another clear advantage of this approach is its significant reduction in both instrumentation and assembly costs.

Some of the advances in the use of neural networks for AO correction can be found in works such as [10,11]. These provide relevant information for further works such as [12], in which convolutional networks are used to study image features by processing them with filters. Another useful work would be [9], where authors use more advanced neural networks like GANs and RNNs with a closed-loop configuration. In this case, GANs provide extra data to the experiment by recreating it based on the learned features.

Building upon the foundation established in [13], this article aims to enhance the techniques and models introduced earlier.

Section 2 explains the points considered when generating data through the simulation software 'Soapy', v0.14.0, as well as comprehensive insight into neural networks, the tool used to carry out the server-based predictions. Section 3 mentions the experiments undertaken in [13]; these have served as the basis for the current work. The foundations of the experiments carried out in the present article are also explained. These include, on the one hand, model training using Tensorflow to subsequently evaluate the efficiency of their predictions with data also generated from the "Soapy" simulator, as well as a second series of tests in which the stability of these models is analyzed under variations in some variables. All experiments are carried out using software and do not involve optical bench or laboratory tests. Section 4 clarifies the experimental outcomes.

2. Methods and Materials

2.1. Adaptive Optics

Adaptive optics [14] encompasses a suite of techniques designed to rectify the distortion of light from distant sources in ground-based telescopes. Atmospheric turbulences introduce variations in the refractive index of the atmosphere, altering the trajectory of incoming light. As a result, plane wavefronts transform into aberrated ones, leading to blurred and diminished quality images.

The essence of AO methodologies is to monitor the impact of these turbulence-induced fluctuations on the wavefront and to instigate real-time corrections. Guide stars serve as reference points to characterize these turbulences. This corrective action relies on three pivotal components: a wavefront sensor, typically the Shack–Hartmann wavefront sensor; a deformable mirror; and a controller or reconstructor. The SH-WFS, comprising a lens array, gauges the local inclinations of the wavefront and relays these data to the reconstructor.

The reconstructor interprets this information and dispatches corrective instructions to the DM actuators. The DM then adjusts its surface in response to these directives to effectuate the required correction. A representative AO system is depicted in Figure 1.

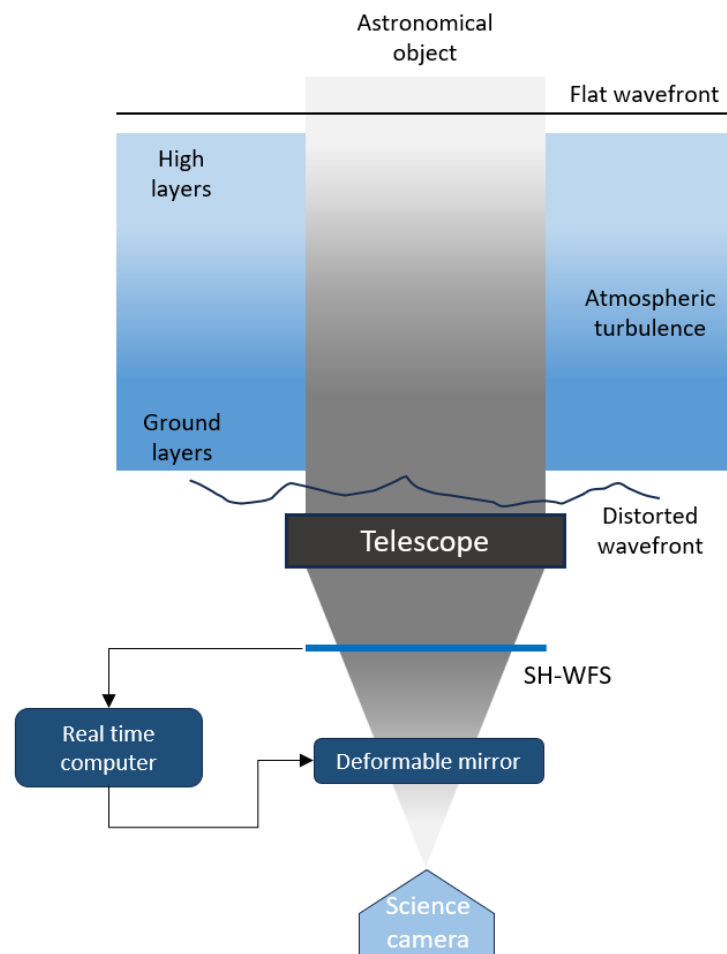


Figure 1. Scheme of single-conjugated AO in open-loop configuration. Distorted light passes through the Shack–Hartmann WFS, which sends information to DM to correct it.

Data used along this paper are provided by a simulation tool, Simulation ‘Optique Adaptative’ with Python (SOAPY) [15]. SOAPY conducts end-to-end simulations of comprehensive AO systems, spanning the creation of atmospheric screens to the final measurements of the telescope’s corrected wavefront. This simulator incorporates a multitude of parameters that allow for a detailed representation of various AO system modules.

As demonstrated in previous experiments [13], the use of slopes as input for neural networks offers a more straightforward option to compute wavefront error predictions, due to the data structure being represented as a numerical matrix, comprising 30 time steps and 72 slopes. Since there are 36 active sub-apertures, the 72 values are divided into 36 for the orthogonal “X” direction and another 36 for the “Y” direction. Once the slopes are obtained, they are processed through the transformation matrix to obtain the Zernike coefficients, which will be used to calculate the wavefront error (WFE) through the interaction matrix.

In a practical case, the predicted slopes would be passed through the wavefront reconstruction algorithm, and a complete representation of the wavefront aberration would be obtained. Then, the necessary adjustments would be made to the deformable mirror actuators to correct the turbulence.

The chosen simulation parameters are encapsulated in Table 1. Leveraging this configuration, the simulations mimic the CANARY low-order SCAO mode’s conditions [16], mirroring the conditions outlined in [13].

Table 1. Main set of parameters for the Soapy SCAO simulation. Unless specified, simulations run with this set of parameters.

Module	Parameter	Value
System	Frequency	150 Hz
	Throughput	1
	Gain	1
Atmosphere	No. phase screens	1
	Wind speeds	from 10 to 15 m/s
	Wind direction	0–360 deg
	Screen height	from 1 to 11 km
	$r_0@500$ nm	0.16 m
	L_0	25 m
Telescope	Diameter	4.2 m
	Central obscuration	1.2 m
SH-WFS	GS magnitude	10
	No. sub-apertures	7×7
	Readout noise	$1 e^-$ RMS
	Photon noise	True
	Wavelength	600 nm
	Thresholding value	0.1
Piezo DM	No. actuators	8×8

A single layer was chosen to mirror the approach taken in [13], where these parameters were used in a similar manner. Essentially, as was mentioned before, the main goal of this study is to build a neural network that produces better predictions by sticking to the original conditions, enabling a comparison between the original model and those developed in the actual work.

On the other hand, opting for a single turbulence layer simplifies the model training process, resulting in a notable reduction in computational costs. Moreover, ref. [13] also demonstrated that using a single layer also obtains good results for scenarios with more than one layer. Furthermore, introducing variability to the ensemble would enhance predictions in such scenarios.

2.2. Neural Networks

Neural networks, inspired by the human cerebral cortex, consist of layered artificial neurons. In 2006, Geoffrey Hinton et al. [17] showcased the significant ability of a neural network to accurately recognize digits. The perceptron, the foundational model of neural networks, traces its mathematical lineage to the work of Warren McCulloch and Walter Pitts in 1943 [18]. This concept set the stage for Frank Rosenblatt’s further development of the perceptron in 1958 [19].

A typical neural network architecture includes an input layer, multiple hidden layers, and an output layer. Neurons in the hidden layers take weighted inputs from the previous layer, and an activation function processes these inputs, allowing the network to map intricate relationships in the data to produce the neuron’s output [20]. In supervised learning, the focus of this study, a labeled dataset is used. It enables the network to adjust its weights and biases iteratively, aiming to minimize the difference between predicted and actual outputs [21]. In the realm of adaptive optics, a great example of neural networks in action can be found as mentioned before in Osborn et al., 2012 [10], where they employed these networks for wavefront reconstruction.

Recurrent neural networks (RNNs) are a specialized subset of deep learning models tailored for sequential data patterns. Unlike traditional neural networks, RNNs have feedback loops, allowing them to disseminate information in a bidirectional temporal manner. This feature makes them apt for time-sensitive data, like atmospheric turbulences. Among RNN variants, long short-term memory (LSTM) is noteworthy. Introduced by Hochreiter and Schmidhuber in 1997 [22], LSTMs address the challenge of vanishing

gradients in training deep networks over extended sequences [23]. LSTMs stand out due to their gating mechanisms, determining which information to retain or discard, emphasizing pivotal details in sequences. Klaus Greff's paper [24] offers an in-depth exploration of enhanced RNNs, juxtaposing them with other neural architectures. Beyond the highlighted benefits, RNNs excel in diverse tasks, such as generating image captions [25] and producing high-quality audio synthesis like WaveNet [26].

Regarding atmospheric turbulence wavefront prediction, ref. [13] emphasizes two benefits of LSTMs:

- **Autonomous learning:** The system does not assume any prior atmospheric knowledge, eliminating the need for manual input during its deployment.
- **Self-tuning:** Dynamic memory elements enable the network to assimilate varying turbulence behaviors autonomously, enhancing its resilience in managing dynamic turbulence aspects.

This study incorporates not just the LSTM but also the 2D-LSTM [27], designed to overcome LSTM limitations by adeptly processing bidimensional sequences. It discerns both spatial and temporal patterns simultaneously, with temporal cells managing time-related nuances and convolutional operations capturing spatial correlations.

3. Experiments

3.1. Prior Experiments

Building on the findings presented in [13], this study delves deeper into the utilization of artificial neural networks for predicting wavefronts. The referenced article examined ANNs in the context of numerical simulations, employing a Shack–Hartmann wavefront sensor [28] in an open-loop configuration. The role of the ANN predictor is to anticipate the uncorrected wavefront slopes for subsequent time steps, basing its forecasts on a series of preceding noisy slope readings. The simulation not only furnishes the predictor with training data but also serves as a platform to gauge its efficacy within the adaptive optics correction mechanism.

After training, an evaluation is carried out on predicted slopes in order to quantify the performance of these models. In that article, three control loop scenarios with and without servo-lag and with the ANN control loop are used, and a comparison of AO corrections in terms of root-mean-square wavefront errors (RMS WFE) is made.

- **Zero delay frame:** Zero-delay or delay-compensated loop, using the current measurement (s_t) immediately.
- **1-Delay frame:** One-frame delay loop, using the prior measurement (s_{t-1}).
- **Predicted frame:** ANN predictive loop, employing the predicted current measurement (s_t) based on ($s_1, s_2, \dots, s_{t-1}$) as input.

The investigation focused on a system simulated with a 7×7 sub-aperture SH-WFS, evaluating the ANN predictor under diverse conditions. Important findings from the original research [13] include the following:

- The ANN predictor markedly enhanced wavefront predictions for the SCAO system operating with a one-frame delay, independent of guide star brightness and wind speeds.
- The ANN predictor showcased resilience against fluctuations on sub-second scales. While the model can be adapted to two-frame delay systems, the results were less than optimal.
- Despite being trained on a solitary atmospheric turbulence layer, the ANN predictor demonstrated proficiency in forecasting wavefronts even under intricate multilayered conditions with distinct wind vectors, albeit with a decline in performance.

3.2. Baseline for Comparison

The conditions delineated in the original study will serve as the initial benchmark, juxtaposing the wavefront error across the three aforementioned scenarios (Figure 2).

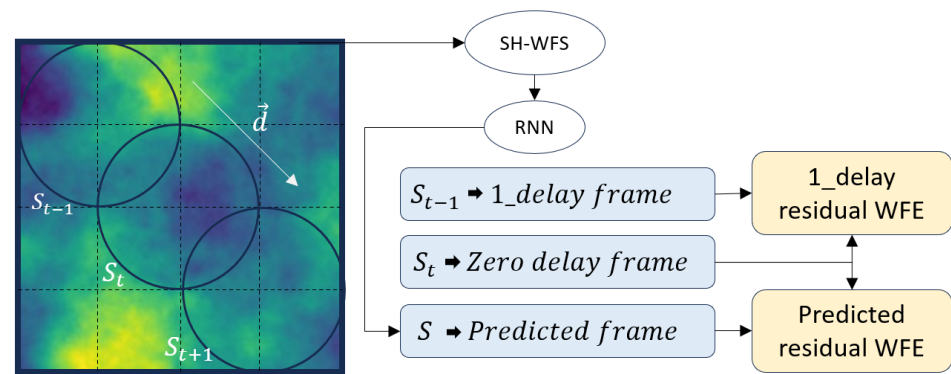


Figure 2. Simulated SCAO system and its data flow. Three cases of slopes are considered for calculating wavefront error in experiments.

This methodology facilitates a nuanced appraisal of the suggested model’s efficacy within a comparable framework. The objective is to scrutinize how various prediction models fare concerning wavefront error prediction via artificial neural networks. However, a vital caveat to bear in mind is the intricacy of mirroring the exact atmospheric conditions delineated in the original research. While most simulation parameters are well defined, certain elements like atmospheric turbulence are complex as they involve both fractal and stochastic elements, posing challenges for precise replication.

To address this limitation and ensure meaningful comparisons, an alternative approach will be adopted to evaluate the accuracy of the proposed models. For model training, root-mean-squared error (RMSE) between the predicted slopes and the real values is used, which implies it is not trained with a cost function dependent on the optical error. However, when performing the shown tests, the calculation of the wavefront error (WFE) is performed, being the percentage of residual error.

Specifically, the error obtained from specific centroids in each of the considered conditions will be compared to the overall error of the system. Calculation of the wavefront error can be found in [13].

$$\text{Residual WFE [\%]}_{\text{delayed}} = \left[\frac{WFE_{\text{zero-delay}} - WFE_{\text{delayed}}}{WFE_{\text{zero-delay}}} \right] \times 100 \quad (1)$$

$$\text{Residual WFE [\%]}_{\text{predicted}} = \left[\frac{WFE_{\text{zero-delay}} - WFE_{\text{predicted}}}{WFE_{\text{zero-delay}}} \right] \times 100 \quad (2)$$

To accomplish this, the results of the previous experiments will be replicated using the same model but with generated data. By doing so, a consistent baseline for comparison will be created, allowing for an assessment of the effectiveness of the novel ANN predictor in a manner that accounts for variations in the atmospheric conditions.

3.3. Models Used in This Work

Derived from the atmospheric conditions and other parameters detailed in Table 1, the dataset comprises 100,000 samples. Each sample encapsulates 100 time steps, and within every step, 72 values depict the slopes or centroids.

For training purposes, a consistent input sequence of 30 time steps is employed. Mirroring the methodology of [13], atmospheric conditions remain unchanged within each sample. The dataset is further augmented by reversing each sequence, emulating the reversal of wind direction. This not only doubles the dataset to 200,000 samples but also enhances model robustness by introducing greater variability in the training data.

Each instance of the datasets (both training and test) is configured with a different height and direction (Table 1), that is, each one is generated with a random height and direction parameter that propagates throughout the 30-frame temporal sequence.

Earlier experiments used a straightforward data structure: the input “x” contained sequences with 30 time steps, each with 72 features, and the output “y” predicted the succeeding time step post the 30-step sequence from “x”. This study, however, has ventured into a novel data arrangement, identified through iterative experimentation to better capture temporal intricacies.

For clarity, consider the matrices representing the training subsets: each matrix corresponds to a dataset sample and a specific slopes value. This structure enables the LSTM model to deeply grasp underlying patterns within temporal data, enhancing its predictive precision. For inputs, the matrices depict the 30-time step sequence ($s_0, s_1, \dots, s_{28}, s_{29}$). Outputs, on the other hand, are delineated by a 2×2 matrix containing two distinct sequences. The first row contains the elements s_{28} and s_{29} , while the second row contains s_{29} and s_{30} , with s_{30} being the time step to be predicted.

$$x_{train} = [s_0 \quad s_1 \quad \dots \quad s_{29}]$$

$$y_{train} = \begin{bmatrix} s_{28} & s_{29} \\ s_{29} & s_{30} \end{bmatrix}$$

The architecture’s design facilitates enhanced interrelation discovery during gradient propagation; in this work it will be called “gradient propagation enhance” (GPE), with an emphasis on augmenting prediction accuracy over extended sequences. The first improvement involves implementing the GPE strategy within the original model, which means retaining the original model as specified in Section 3.2.2 and Table 2 in [13]. This results in what we refer to as the “GPE model”. As the main goal of this research is to improve the model’s ability to make accurate predictions for multiple time steps, not just the first one, it was decided to completely overhaul the model and its structure, with the goal of achieving significant improvements in both short-term and long-term predictions. No longer limited to basic LSTMs, it now integrates multiple layers, inclusive of 2D-LSTMs and 2D convolutional layers. This diversification enables the model to evaluate both temporal and spatial relationships, refining predictive accuracy. As in the previous case, this model will be referred to as the “2D-LSTM model” in future experiments.

The neural network’s specifics can be referenced in Table 2. It amalgamates ConvLSTM2D layers for sequential data handling, BatchNormalization layers for normalization, and Conv2D layers for 2D convolutional operations. Together, these layers analyze input data to produce the final output: a prediction founded on the processed time step sequence. The network boasts 2,365,465 parameters, fine-tuned during training to optimize its efficacy.

Table 2. The network is depicted, showing the layers that compose it, along with the input and output sizes, as well as the trainable parameters comprising the network.

Layer	Input Shape	Output Shape	Parameters
InputLayer	(30, 1, 72, 1)	(30, 1, 72, 1)	
ConvLSTM2D	(30, 1, 72, 1)	(30, 1, 72, 128)	594,944
BatchNormalization	(30, 1, 72, 128)	(30, 1, 72, 128)	512
ConvLSTM2D	(30, 1, 72, 128)	(30, 1, 72, 128)	1,180,160
BatchNormalization	(30, 1, 72, 128)	(30, 1, 72, 128)	512
ConvLSTM2D	(30, 1, 72, 128)	(1, 72, 128)	1,180,160
BatchNormalization	(1, 72, 128)	(1, 72, 128)	512
Conv2D	(1, 72, 128)	(1, 72, 1)	1153
BatchNormalization	(1, 72, 1)	(1, 72, 1)	4
Conv2D	(1, 72, 1)	(1, 72, 1)	10

In this scenario, specifying the number of instances for the input layer is not necessary. This is because implementations of predefined models of recurrent neural networks in libraries like Keras and TensorFlow allow flexibility in the temporal dimension, and it would be determined automatically at the time of model training.

Importantly, the need for custom layers is obviated thanks to the availability of both LSTM-2D layers and others in the Keras library [29]. An alternate model was probed in [30], where an LSTM encoder–decoder–predictor model was designed to concurrently reconstruct the input sequence and predict future sequences. However, their model’s fully connected LSTM layer overlooked spatial correlation, rendering the 2D-LSTM approach more favorable.

3.4. ANN Training and Optimization

The dataset is partitioned such that 90% of instances are earmarked for network training, while the remaining 10% is set aside for validation. Notably, regularization techniques like Dropout were abstained from being employed, given that performance has not been notably enhanced post model convergence. The mean-squared error (MSE) was chosen as the loss function, which quantifies the average squared discrepancy between actual and predicted outputs. The choice of the Adam optimizer stems from its efficacy in training models on sequential data. Adam blends the adaptive gradient algorithm with momentum-based optimization, adeptly managing diverse gradients [31]. Its auto-adaptive learning rates, informed by gradient history, tackle issues of vanishing/exploding gradients. The resultant momentum, combined with adaptive learning rates, accelerates training, culminating in more refined convergence for intricate tasks.

To ensure proper convergence, training process is initiated with Adam’s default learning rate which is 1×10^{-3} , and incorporates a reduction strategy using the “ReduceLROnPlateau” [32] function with a factor of 1/5 and a patience of 5 epochs. The training is configured for 40 epochs, and optimal results are typically observed between epochs 30 and 40, with variations depending on the predicted time step.

Delving into the specifics of the LSTM-2D layers, each layer employs 128 filters, paired with “same” padding and a 3×3 kernel size. The “return sequences” attribute is activated (set to “True”) for the initial two layers, ensuring each layer reciprocates with a sequence of outputs corresponding to every time step in the input sequence. However, the final layer modifies this parameter to “False”, returning only the ultimate output after processing the entire input sequence. As for the Conv2D layers, they mirror the padding and kernel size of their LSTM-2D counterparts. A singular filter is employed, advocating for a streamlined architecture with reduced parameters. This minimizes complexity and diminishes the potential for overfitting.

3.5. Experiments Conducted in This Work

The experiments can be bifurcated into two categories. First, the primary objective is to identify models that outperform previously identified ones. Concurrently, the chosen model’s performance and stability across a gamut of scenarios are assessed.

Initially, the model structure mirroring the original study is utilized under the GPE framework. Subsequent iterations aim to leverage the LSTM-2D alongside convolutional layers, with a goal to improve predictions for not only the immediate but also future time steps. Once these models were in place, a plethora of tests were conducted across varying atmospheric conditions and input data configurations. These served to elucidate the strengths and potential vulnerabilities of the proposed models. Due to computational constraints, testing prioritized the most promising scenarios, focusing on the 2D-LSTM model within the immediate time step framework.

- Sensitivity to noise level: As mentioned previously, in adaptive optics systems, guide stars are used as references to measure aberrations. In our specific case, with an SCAO configuration, we employ a point source at infinity to serve as a natural guide star (GS). That being said, the parameter of the guide star magnitude (GS Mag) refers to the brightness of this star, indicating that the higher the magnitude of the guide star, the fainter the flux from that GS, and so there is more noise [33]. In this experiment, an analysis of the model’s efficiency is conducted under different levels of noise in the WFS, allowing for an assessment of the model’s stability in response to variations

in this parameter. As demonstrated in [13], a model trained on GSMag magnitudes that makes predictions on data of the same magnitude achieves higher performance, especially when dealing with a magnitude value of 10. While this magnitude was incorporated into the training data, the model's stability was further probed across lower magnitudes (ranging from 0 to 8). The intent was to ascertain whether model accuracy remained consistent under these parameters.

- **Analysis of different turbulence strength:** In the upcoming scenario, the focus will be on analyzing the Fried parameter, denoted as r_0 . This parameter is a commonly used metric in atmospheric science, particularly within the realm of astronomical observations, and is usually defined at 500 nm. The model is trained, as previously established, with simulated data considering a r_0 of 16 cm. This value was chosen to imitate the case of the experiments of [13], as was performed in the different experiments of the work. Two extreme cases were considered for the test, with an r_0 of 8 and 30 cm. However, throughout the time sequence, this parameter is kept constant in order to evaluate the performance of the model in the presence of a turbulence force of different magnitude. Theoretically, higher r_0 values should simplify model predictions. A diminished r_0 suggests a stronger optical turbulence, potentially complicating accurate forecasting.
- **Multiple layers of turbulence:** The training model is based on a singular turbulence layer at a random wind direction and height. Yet, real-world scenarios often present multiple atmospheric layers, introducing a complex turbulence matrix. Consequently, the model's resilience in multilayer scenarios is put to the test. Evaluations are segmented into three cases, with 1, 2, 5, 10, and 20 layers, respectively. Only the number of layers is changed; all other parameters in Table 1 remain constant.

4. Results

When visualizing the results, both improvement scenarios will be considered in the following manner. Firstly, the original model will be modified with the new training process, resulting in the creation of the GPE model, and secondly, the new 2D-LSTM model will be introduced. In both cases, the corresponding delay (s_{29}) will be used as a reference, and the results will be compared between the original model and the new ones. Subsequently, each of the three experiments conducted to assess the model's stability in response to atmospheric parameter modifications will be presented sequentially. These experiments include sensitivity to noise levels, turbulence strength r_0 , and the case of an atmosphere composed of multiple turbulence layers, in that order.

In the process of testing prediction models, 2000 instances will be considered. However, numerical variations may be observed in the instances when turbulence parameters are adjusted, as was the case in the last three experiments. Each instance comprises a sequence of 100 frames, and this data sequence will be processed to generate predictions for each corresponding time step, resulting in a total of 100 predictions. To ensure clarity and to steer clear of regions where the models have not yet stabilized due to insufficient input data, the results are presented starting from frame number 40 onwards.

The evaluation results will be presented using two terms: "average wavefront residual error" (AvgWFER) and "error reduction" (ERr):

- **Average wavefront residual error:** It is derived from the average of the residual errors for each of the 100 predictions in the sequence, representing the mean value for each prediction across the 2000 instances under consideration.
- **Error reduction:** Based on the AvgWFER of the original models, GPE, 2D-LSTM, and the corresponding AvgWFER for each case, such as the 1-delay case for future time step prediction (s_{30}), the error reduction achieved with each modification is calculated as a percentage.

Within the experiment charts, one can observe a shaded area surrounding the curve, representing the standard deviation of the 2000 instances considered for each prediction in the sequence. This shading provides a visual representation of the variability associated with the predictions.

To conduct an analysis of the models' predictions and their performance, we focus on three specific time steps, denoted as the "First", "Second", and "Third time step". As mentioned before, a sequence of 30 time steps is considered in training, from s_0 to s_{29} . That is, the "First time step" corresponds to step s_{30} , the "Second time step" to s_{31} , and the "Third time step" to s_{32} . Subsequent time steps are excluded from consideration given the random nature of the atmospheric turbulence.

4.1. GPE Model

In the case of GPE, only the results for the first and second time steps are displayed, as there is minimal improvement observed in the third one compared to the model developed in the original study. Figures 3 and 4 show the performance of model GPE compared to the other cases, while Tables 3 and 4 show the average residual error values, as well as the error reduction from the 1-delay case reference in %, for the first and second frame cases, respectively.

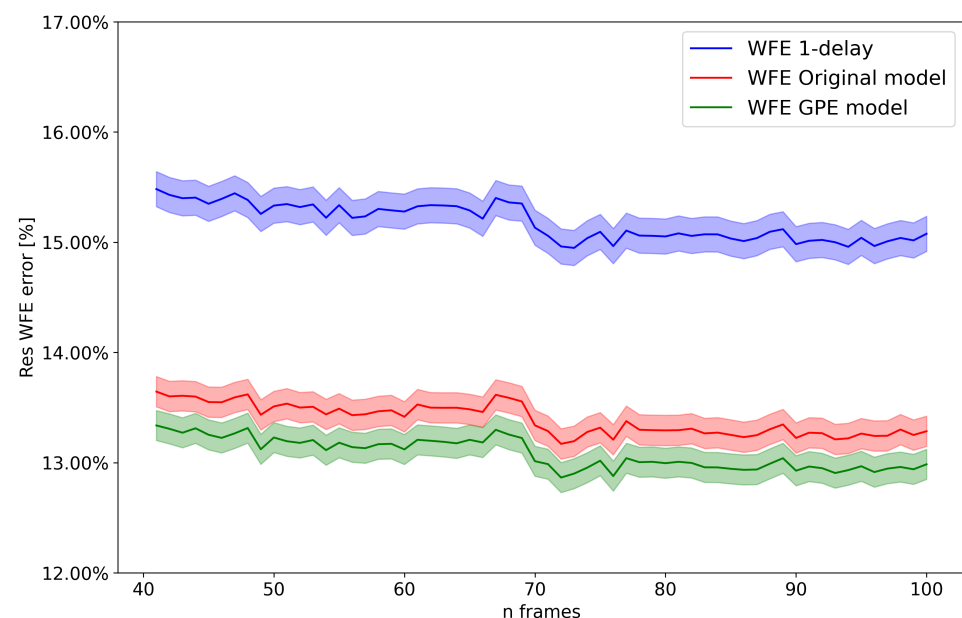


Figure 3. Residual error for 1-delay, original model, and GPE model. First frame.

Table 3. First time step average residual error and proper reduction with original and GPE models.

Case	Avg Res Error [%]	Error Reduction [%]
1-delay	15.21	Ref
Original model	13.42	11.76
GPE model	13.11	13.81

Table 4. Second time step average residual error and proper reduction with original and GPE models.

Case	Avg Res Error [%]	Error Reduction [%]
2-delay	24.88	Ref
Original model	19.93	19.89
GPE model	19.72	20.74

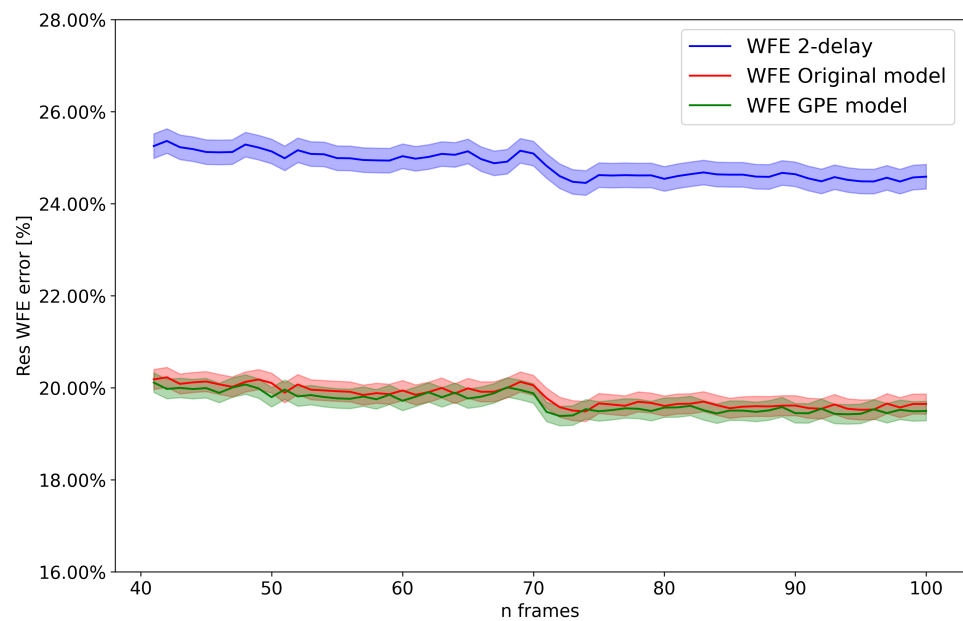


Figure 4. Residual error for 2-delay, original, and GPE models. Second frame.

4.2. 2D-LSTM model.

In contrast, when evaluating the 2D-LSTM architecture, as mentioned earlier, the most favorable real-time predictions are observed under the conditions and telescope configuration assumed within the simulation. This enhancement is evident in both error reduction and the capacity to forecast further into the future, offering the potential for in-depth analysis in more advanced time steps beyond the initial three considered.

Figures 5–7 depict a scenario similar to the one discussed in the GPE case. Here, the results of the original model are being compared with those from the 2D-LSTM structure. The examination of events during the third time step is also being carried out due to observed improvement.

Tables 5–7 show the average residual error values, along with the corresponding error reduction percentages compared to the 1-delay case reference. These values pertain to the first, second, and third frame cases, respectively.

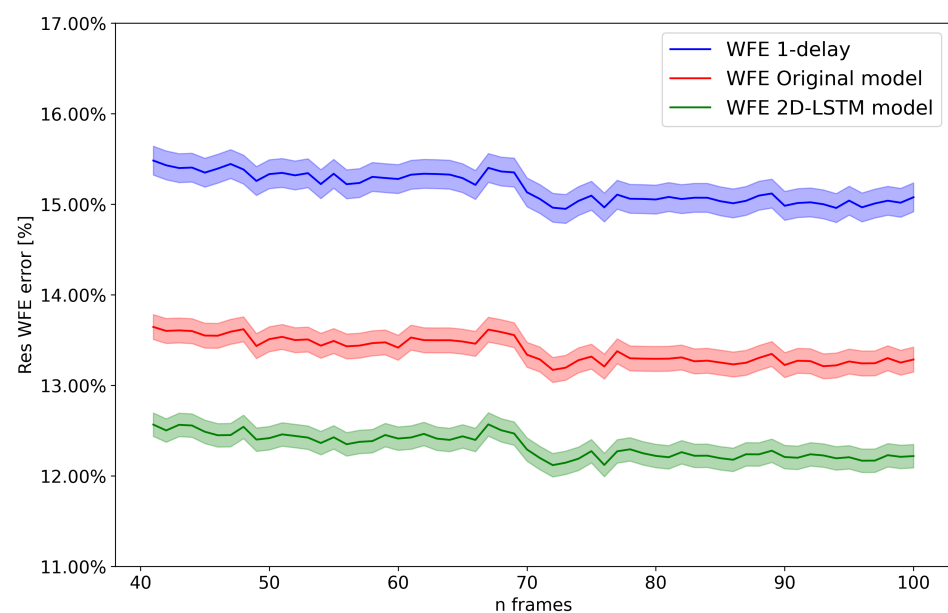


Figure 5. Residual error for 1-delay, original, and 2D-LSTM models. First frame.

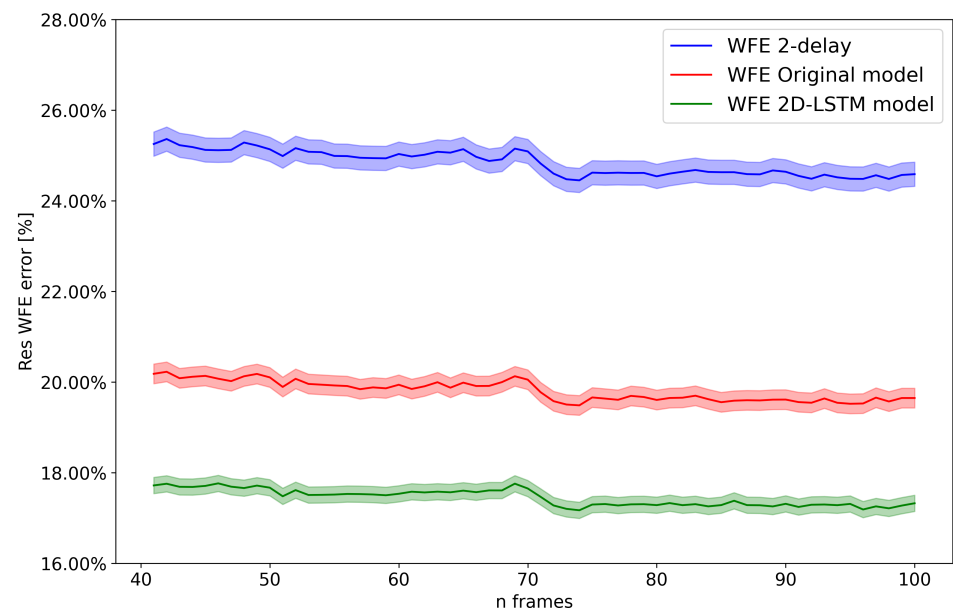


Figure 6. Residual error for 2-delay, original, and 2D-LSTM models. Second frame.

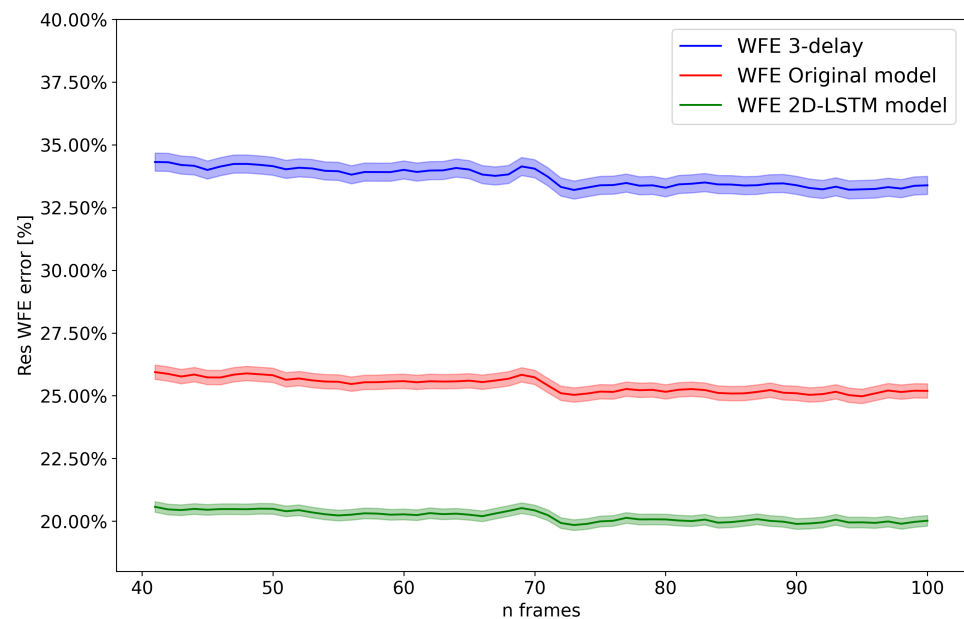


Figure 7. Residual error for 3-delay, original, and 2D-LSTM models. Third frame.

Table 5. First time step average residual error and proper reduction with original and 2D-LSTM models.

Case	Avg Res Error [%]	Error Reduction [%]
1-delay	15.21	Ref
Original model	13.42	11.76
2D-LSTM model	12.35	18.63

Table 6. Second time step average residual error and proper reduction with original and 2D-LSTM models.

Case	Avg Res Error [%]	Error Reduction [%]
2-delay	24.88	Ref
Original model	19.93	19.89
2D-LSTM model	17.77	28.58

Table 7. Third time step average residual error and proper reduction with original and 2D-LSTM models.

Case	Avg Res Error [%]	Error Reduction [%]
3-delay	33.77	Ref
Original model	25.47	24.57
2D-LSTM model	20.50	39.29

4.3. Developed Models Comparison

After performing an analysis of both models, for the subsequent two time steps in the case of the GPE (since the third one has hardly changed) and three time steps for the 2D-LSTM, it is clear that the latter model outperforms the former. While GPE showcases a short-term enhancement comparing to the original model, the 2D-LSTM allows this improvement to propagate to time steps further in the future. For an overview of the results, refer to Tables 8 and 9.

Table 8. First time step average residual error and proper reduction, global comparison.

Case	Avg Res Error [%]	Error Reduction [%]
1-delay	15.21	Ref
Original model	13.42	11.76
GPE model	13.11	13.81
2D-LSTM model	12.35	18.63

Table 9. Second time step average residual error and proper reduction, global comparison.

Case	Avg Res Error [%]	Error Reduction [%]
2-delay	24.88	Ref
Original model	19.93	19.89
GPE model	19.72	20.74
2D-LSTM model	17.77	28.58

4.4. Sensitivity to Noise Level

It is expected that when training the model with a GSMag of magnitude 10, a good prediction can be achieved for data with that magnitude. However, the experiment demonstrates that, when data with a lower magnitude is used, and, therefore, there is less noise in the sensor, predictions of similar performance are still obtained. In Figure 8, the comparison between the residual errors of each magnitude with respect to the 1-delay time step reference can be observed. Given that each case corresponds to a 1-delay reference, it results in different curve shapes. Nevertheless, graphically, we have omitted all 1-delay cases, except the main reference, due to their relative similarity in terms of the average term. In Table 10, the average residual error values are presented for every GSMag value, along with the corresponding error reduction percentages compared to the 1-delay reference. In this section, numerical results are presented for each of the considered cases.

Table 10. First time step average residual error and proper reduction for 2D-LSTM model with different GSMag.

Case	Avg Res Error [%]	Error Reduction [%]
1-delay (GS-10)	15.33	Ref (GS-10)
1-delay (GS-8)	15.28	Ref (GS-8)
1-delay (GS-4)	15.21	Ref (GS-4)
1-delay (GS-0)	15.32	Ref (GS-0)
GSMag-10	12.43	18.86
GSMag-8	12.36	19.14
GSMag-4	12.11	20.39
GSMag-0	12.21	20.29

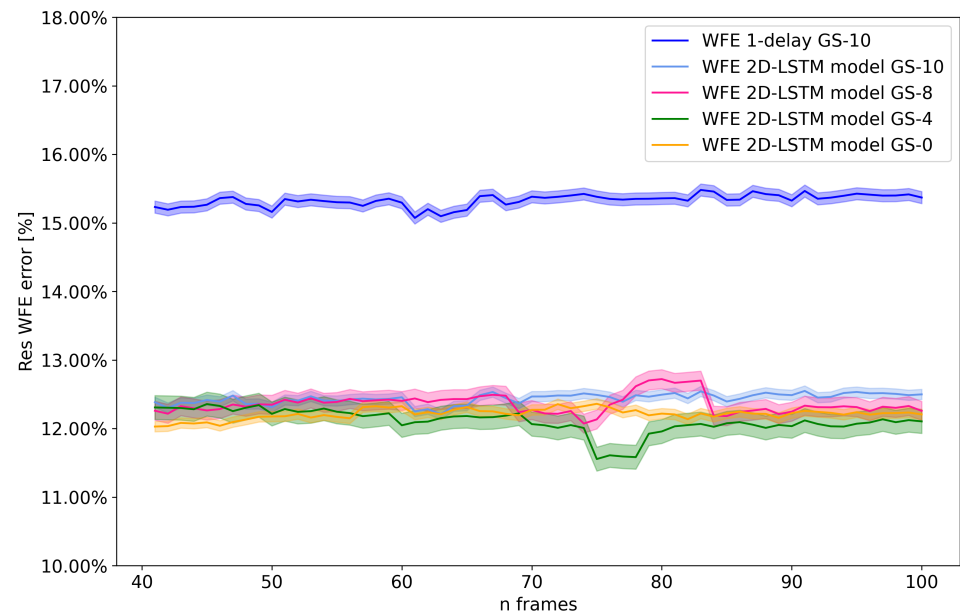


Figure 8. Residual error for 1-delay and 2D-LSTM model with different GSMag values. First frame.

4.5. Analysis of Different Turbulence Strength

As mentioned before, a larger r_0 value indicates a more stable atmosphere, which means fewer fluctuations and distortions in the light passing through it. Conversely, a smaller r_0 value implies a more unstable atmosphere with greater turbulence, making it harder to obtain clear and sharp images.

Based on the tests carried out, when a r_0 magnitude of 8 cm is considered, a smaller error reduction can be observed (in Figure 9) than in the other scenarios, although it is not far away in terms of residual WFE, as it is shown that the model is able to provide good predictions even with a considerably unfavorable r_0 . In the 30 cm scenario, a greater reduction in error is observed compared to the 16cm case for which the model is trained. This is expected since, as mentioned earlier, the conditions are significantly better, making error reduction easier. In Table 11, the average residual error values are presented for each one of the r_0 values, along with the corresponding error reduction percentages compared to their respective 1-delay case references.

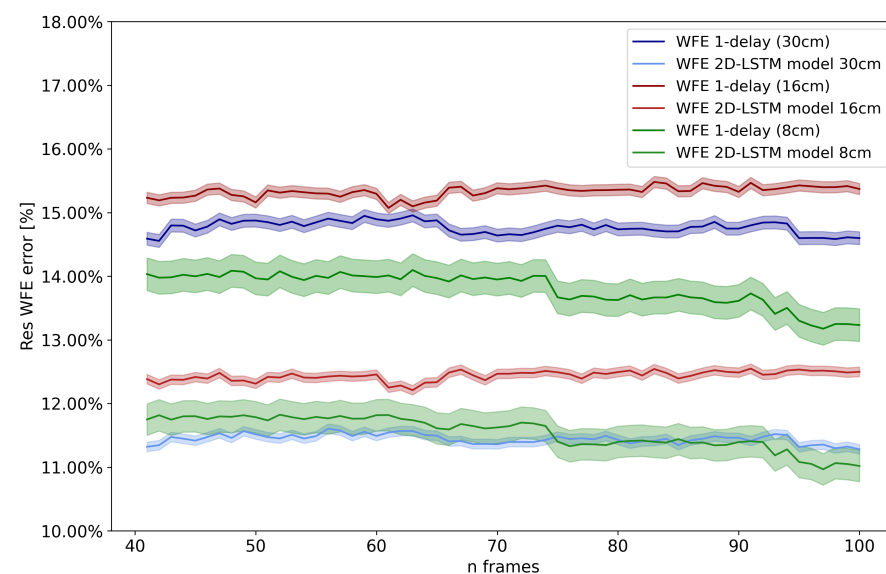


Figure 9. Residual error for 2D-LSTM model with different r_0 values. First frame.

Table 11. First time step average residual error and proper reduction for 2D-LSTM model with different r_0 .

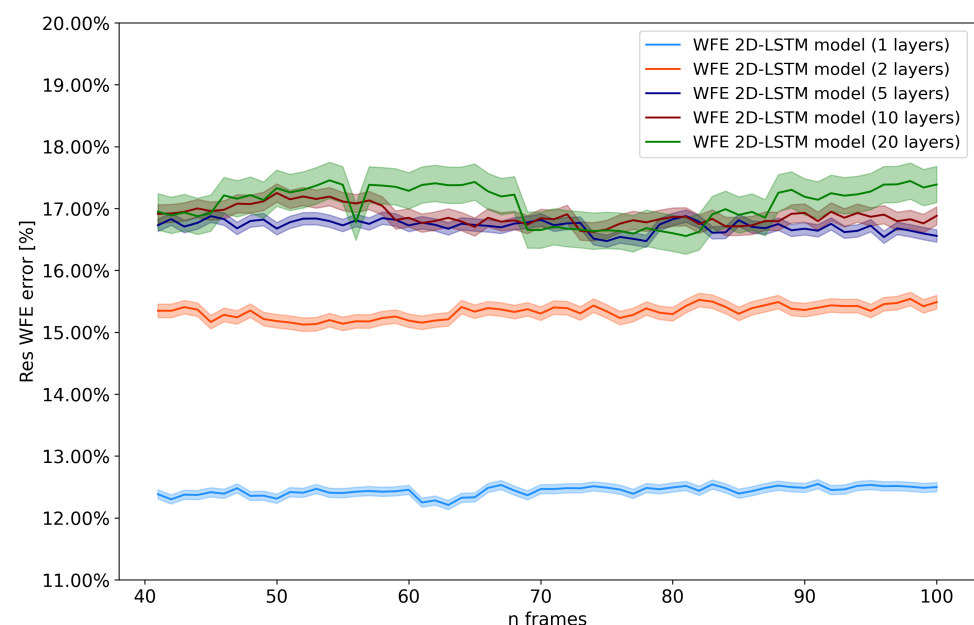
Case	Avg Res Error [%]	Error Reduction [%]
1-delay (30 cm)	14.76	Ref for 30 cm
1-delay (16 cm)	15.33	Ref for 16 cm
1-delay (8 cm)	13.80	Ref for 8 cm
30 cm	11.44	22.49
16 cm	12.43	18.86
8 cm	11.54	16.34

4.6. Multiple Layers of Turbulence

In this experiment, multiple datasets generated by the simulator are analyzed using the parameters outlined in Table 1. However, in this scenario, the modified variable is the number of turbulence layers. The objective is to assess the model's performance when compared with more complex atmospheres. Each instance represents a time sequence with different height and wind direction, providing a comprehensive evaluation of the model's capabilities under diverse conditions.

Each of these turbulence layers represents a different part of the atmosphere where the properties of the air can change considerably. Therefore, it is natural to consider that when more turbulence layers are introduced, the model might not perform as well.

In Figure 10, it is observed that as the number of turbulence layers increases, the model faces greater difficulty in reducing the residual error. However, there is also a significant variation in the residual error for the 1-delay corresponding to each case, as illustrated in Table 12. This table presents the average residual error values for different multilayer scenarios, accompanied by the respective error reduction percentages in comparison to the 1-delay case reference.

**Figure 10.** Residual error for 2D-LSTM model with different multilayer configurations. First frame.**Table 12.** First time step average residual error (reference errors are included) and proper reduction for 2D-LSTM model with different multilayer configurations.

Case	Ref Error [%]	Avg Res Error [%]	Error Reduction [%]
1 layer	15.33	12.43	18.86
2 layers	16.68	15.33	8.11

Table 12. Cont.

Case	Ref Error [%]	Avg Res Error [%]	Error Reduction [%]
5 layers	17.40	16.72	3.91
10 layers	17.41	16.89	2.93
20 layers	17.51	17.08	2.50

4.7. Results Overview

While the modified version of the original model (GPE model) only achieves better short-term performance for the first predicted time step s_{30} , and slightly better for the second time step s_{31} , in the case of the 2D-LSTM, both short-term and long-term improvements are obtained. This is because the second case considers a combination of convolutional layers and two-dimensional LSTMs that allow the capture of both spatial and temporal features, as opposed to the original model in which only the time evolution of the atmospheric turbulence data was considered.

Another point to bear in mind is the fact that the 2D-LSTM model demands more computational resources for training a big amount of data but less for a single sample (as typically performed in real-world scenarios). Furthermore, the smooth implementation of keras and Cuda with this type of neural network layers allows a very efficient integration which results in much more efficient training processes with the use of GPUs.

Additionally, there is a brief period of several milliseconds required to achieve stability, as illustrated in Figures 5–7. However, this short delay is not of significant relevance, given that the system is designed for prolonged operation.

In Sections 4.4–4.6, corresponding evaluations of the trained model against a magnitude change in three different variables are carried out. In the case of the star magnitude, since it was trained with a magnitude of 10, good results are obtained when performing the tests for lower magnitudes with a lower amount of noise, which would be expected. However, it is important to note that the performance obtained by a model trained with GSMag 10, as demonstrated in the original work, offers the best predictions for datasets made up of data simulated with other magnitudes, which can also be observed in the present work. More importantly, with the new 2D-LSTM model, the residual errors for different magnitudes remain more or less in the same range, which means that it performs better than the original model.

In Section 4.5, tests are considered based on the two extreme values of r_0 (8 cm and 30 cm), in which, again, favorable results are found, since the error reduction in the worst of the three cases considered, a magnitude of 8 cm, is achieved down to optimal levels by improving the “1-delay” reference case, the time step s_{29} . As expected, at the value used for training, good results are obtained as in the simpler atmosphere case with an r_0 of 30 cm.

Finally, in Section 4.6, good results are obtained for the prediction of up to five turbulence layers; however, as the number of turbulence layers increases, the performance decreases significantly with respect to the single-layer case. However, despite this loss of efficiency, the residual error is still better than that of the 1-delay reference. Future experiments could consider creating a more varied dataset composed of more layers, so that the effect of data variability in predicting a multilayered atmosphere could be analyzed.

5. Conclusions and Future Lines

The research underscores the potential for enhancing adaptive optics wavefront prediction models. It reaffirms that with the deployment of sophisticated neural network structures, better performance metrics can be achieved than what is presently attainable. Notably, the 2D-LSTM model’s adeptness in processing both temporal and spatial data promotes accurate predictions beyond a single time step. Further analysis could explore the extent of this capability for a greater number of frames, thereby assessing how far it can be extended without compromising the model’s performance compared to the results of the 1-delay frame reference case.

Diverse atmospheric parameters, including sensor noise magnitude and turbulence strength, were tested. While deviations from optimal conditions led to slightly reduced efficacy, the predictions remained within acceptable bounds. Enriching the training data with a wider range of values for these parameters could potentially amplify the model's resilience against environmental fluctuations.

Looking ahead, there are several promising avenues for further research. One of these involves considering the prediction of noise-free data as the output from noisy input data into the network. This approach has the potential to yield clean slopes, thus reducing a portion of the error that might occur in the image correction process.

Another possibility worth exploring is the feasibility of enhancing processes through cascading predictions or similar methods, or potentially partitioning the data. Furthermore, there is a compelling opportunity to study the model's efficiency and optimize it to reduce computational costs while minimizing any compromise in prediction accuracy.

Additionally, there are plans to conduct real-world tests using an optical bench that will enable the replication of the atmospheric conditions considered in this study within the “Soapy” simulator. This will facilitate a direct comparison between results obtained from data generated by a pseudo-random algorithm and real atmospheric turbulence.

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